```python
# ■■ STEP 1: DATA LOADING ■■■
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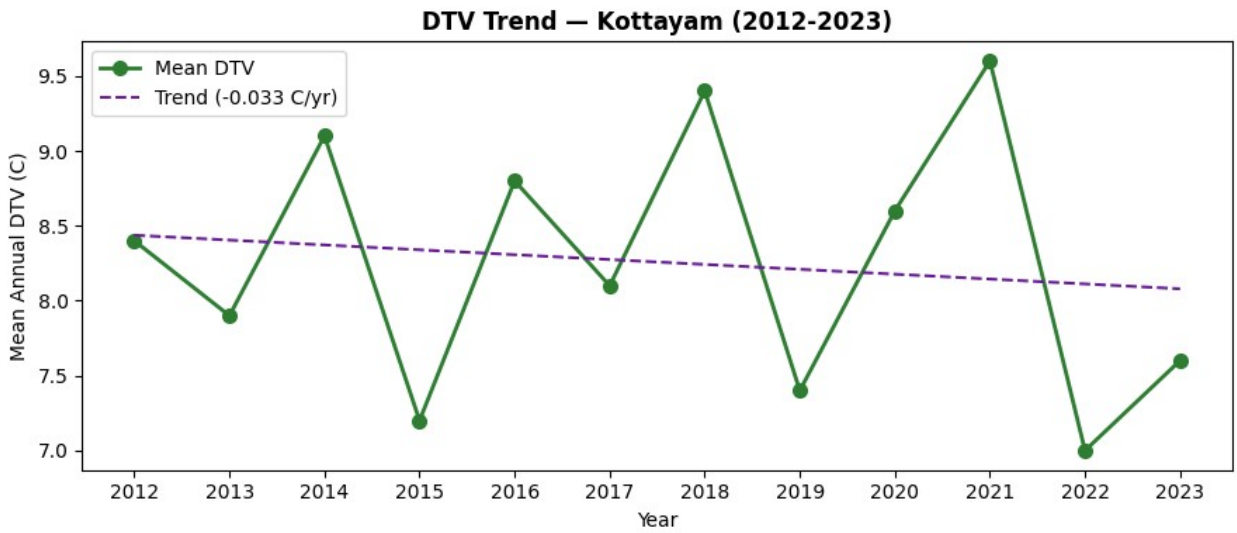
```
preds = np.array(preds)
# ■■ STEP 4: PREDICTION & METRICS ■■■
```

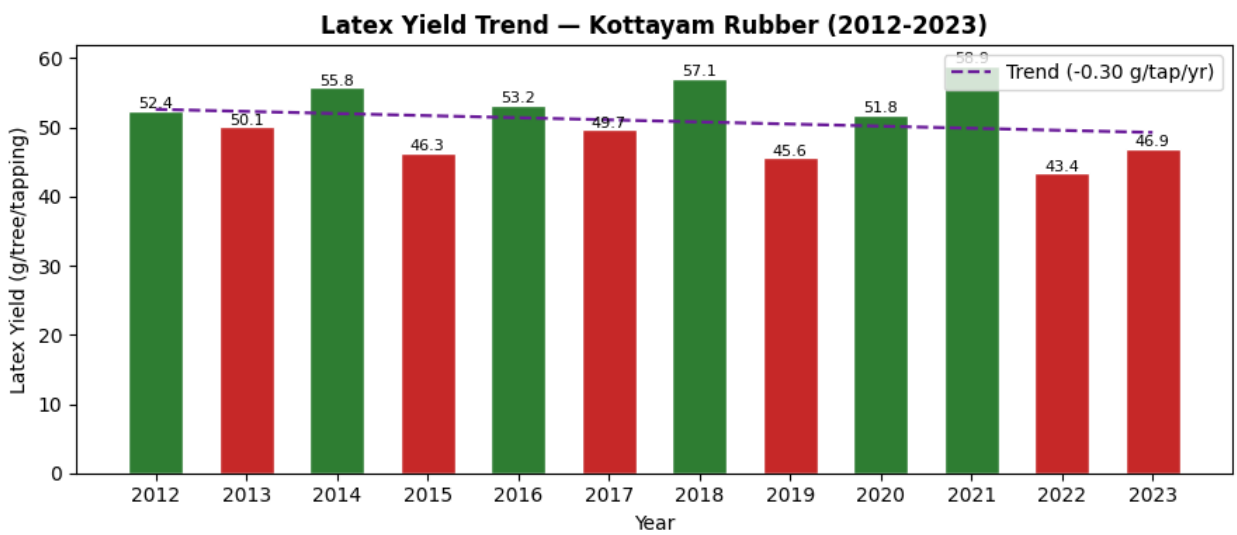
```
bar_c=['#2e7d32','#c62828','#0d47a1','#e65100','#6a1b9a','#37474f']
plt.figure(figsize=(8,4))
plt.bar(range(6),importances[sorted_idx],
color=[bar_c[i] for i in sorted_idx],edgecolor='white')
plt.xticks(range(6),[feat_labels[i] for i in sorted_idx],fontsize=9)
for i,v in enumerate(importances[sorted_idx]):
    plt.text(i,v+0.005,f'{v:.3f}',ha='center',fontsize=9)
plt.title('RF Feature Importance — Latex Yield
Prediction',fontweight='bold')
plt.ylabel('Importance Score (Gini)')
plt.tight_layout(); plt.savefig('fig3_importance.png',dpi=150);
plt.show()
# ■■ GRAPH 4: Predicted vs Actual ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
plt.figure(figsize=(9,4))
plt.plot(years,y,'o-',color='#2e7d32',lw=2.2,ms=7,label='Actual
Yield')
plt.plot(years,preds,'s--',color='#c62828',lw=2,ms=6,
label='RF Predicted (LOO-CV)')
plt.fill_between(years,y,preds,alpha=0.15,color='#6a1b9a',label='Error
band')
plt.text(0.5,0.05,f'R2={r2:.3f} | MAE={mae:.1f} | RMSE={rmse:.1f}',
transform=plt.gca().transAxes,ha='center',fontsize=9,
bbox=dict(boxstyle='round',facecolor='white',alpha=0.9))
plt.title('RF: Actual vs. Predicted Latex Yield (LOO-
CV)',fontweight='bold')
plt.xlabel('Year'); plt.ylabel('Latex Yield (g/tree/tapping)')
plt.xticks(years); plt.legend()
plt.tight_layout(); plt.savefig('fig4_actual_vs_pred.png',dpi=150);
plt.show()
# ■■ GRAPH 5: Correlation Heatmap (Seaborn) ■■■■■■■■■■■■■■■■■■■■
plt.figure(figsize=(7,6))
corr=df[['DTV','T_min','Rainfall','Tapping_Days','HSI','Latex_Yield']]
.corr()
sns.heatmap(corr,annot=True,fmt='.2f',cmap='RdYlGn',
vmin=-1,vmax=1,linewidths=0.5,
xticklabels=['DTV','T-min','Rain','Tap Days','HSI','Yield'],
yticklabels=['DTV','T-min','Rain','Tap Days','HSI','Yield'])
plt.title('Pearson Correlation Heatmap',fontweight='bold')
plt.tight_layout(); plt.savefig('fig5_heatmap.png',dpi=150);
plt.show()
# ■■ GRAPH 6: Residual Plot ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
residuals = y - preds
plt.figure(figsize=(9,4))
bar_r=['#2e7d32' if r>=0 else '#c62828' for r in residuals]
plt.bar(years,residuals,color=bar_r,width=0.6,edgecolor='white')
plt.axhline(0,color='black',lw=1.2,ls='--')
plt.title('Model Residuals — RF LOO-CV (Actual —
Predicted)',fontweight='bold')
plt.xlabel('Year'); plt.ylabel('Residual (g/tree/tapping)')
plt.xticks(years,rotation=45)
```

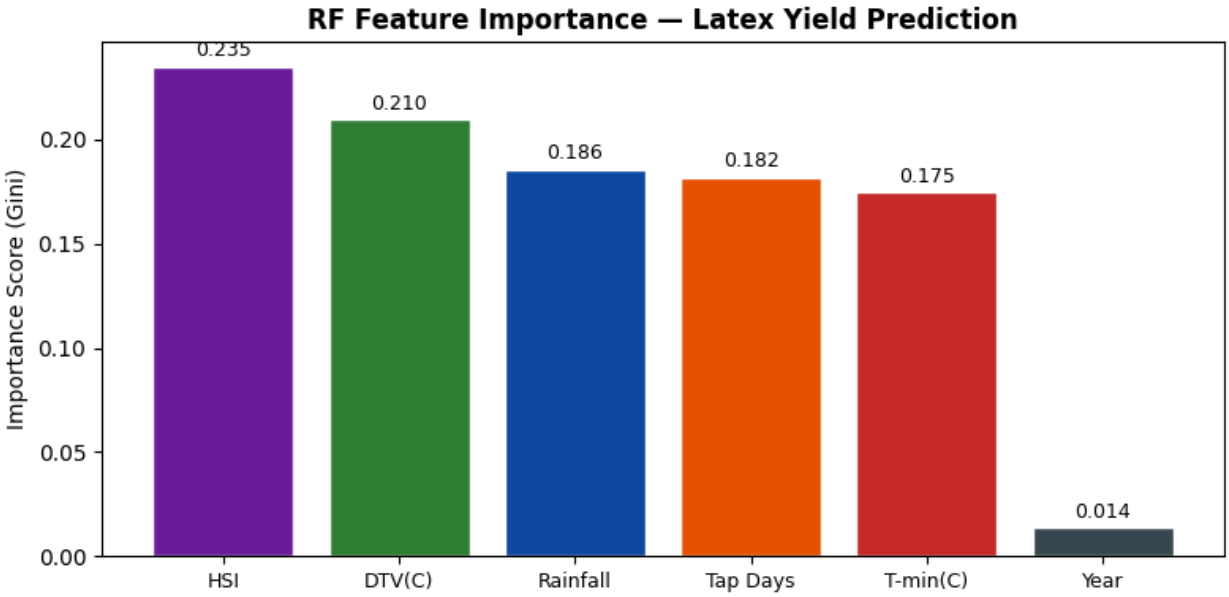
```
plt.tight_layout(); plt.savefig('fig6_residuals.png',dpi=150);
plt.show()

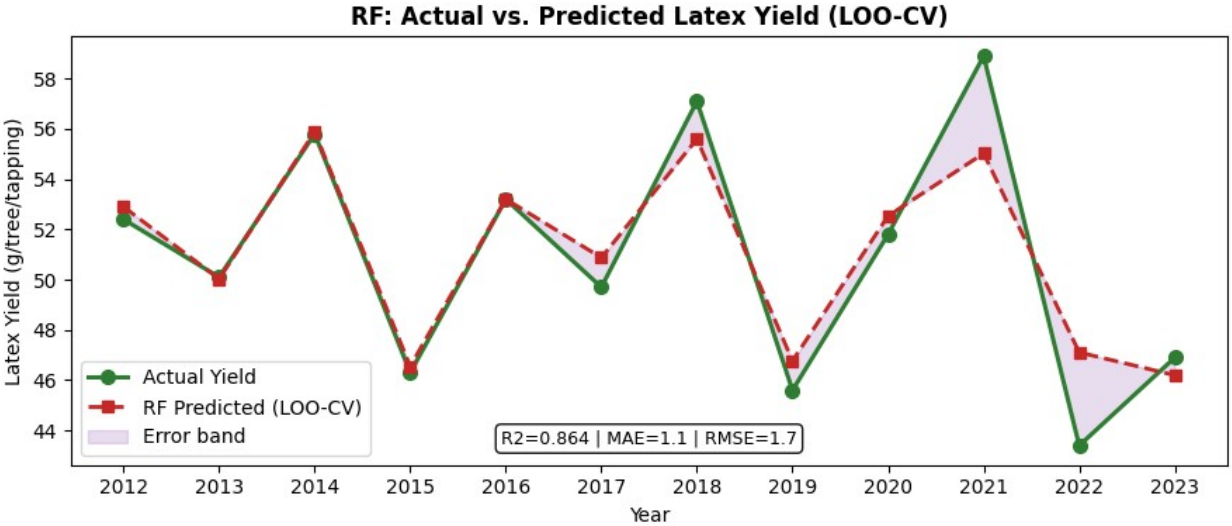
Dataset loaded:
 Year  DTV  T_min  Rainfall  Tapping_Days  HSI  Latex_Yield
 2012  8.4   21.8      2760           218   42         52.4
 2013  7.9   22.1      2940           209   51         50.1
 2014  9.1   21.4      2620           226   38         55.8
 2015  7.2   22.8      3210           194   55         46.3
 2016  8.8   21.9      2710           221   44         53.2
 2017  8.1   22.4      2880           210   48         49.7
 2018  9.4   21.2      2550           230   35         57.1
 2019  7.4   23.1      3180           197   58         45.6
 2020  8.6   22.2      2790           216   46         51.8
 2021  9.6   20.9      2490           234   31         58.9
 2022  7.0   23.4      3350           188   61         43.4
 2023  7.6   23.2      3090           199   57         46.9
          Year    DTV  T_min  Rainfall  Tapping_Days    HSI
Latex_Yield
count    12.00  12.00  12.00     12.00         12.00  12.00
12.00
mean   2017.50   8.26  22.20   2880.83        211.83  47.17
50.93
std       3.61   0.87   0.81    277.57         14.90   9.60
4.85
min    2012.00   7.00  20.90   2490.00        188.00  31.00
43.40
25%    2014.75   7.55  21.70   2687.50        198.50  41.00
46.75
50%    2017.50   8.25  22.15   2835.00        213.00  47.00
50.95
75%    2020.25   8.88  22.88   3112.50        222.25  55.50
53.85
max    2023.00   9.60  23.40   3350.00        234.00  61.00
58.90
Feature matrix shape: (12, 6)
Target vector shape:  (12,)
R2 = 0.864
MAE = 1.15 g/tree/tapping
RMSE= 1.71 g/tree/tapping
Year | Actual | Predicted | Error
2012 52.4 52.9 0.5
2013 50.1 50.0 0.1
2014 55.8 55.9 0.1
2015 46.3 46.6 0.3
2016 53.2 53.2 0.0
2017 49.7 50.9 1.2
2018 57.1 55.6 1.5
2019 45.6 46.8 1.2
2020 51.8 52.5 0.7
```

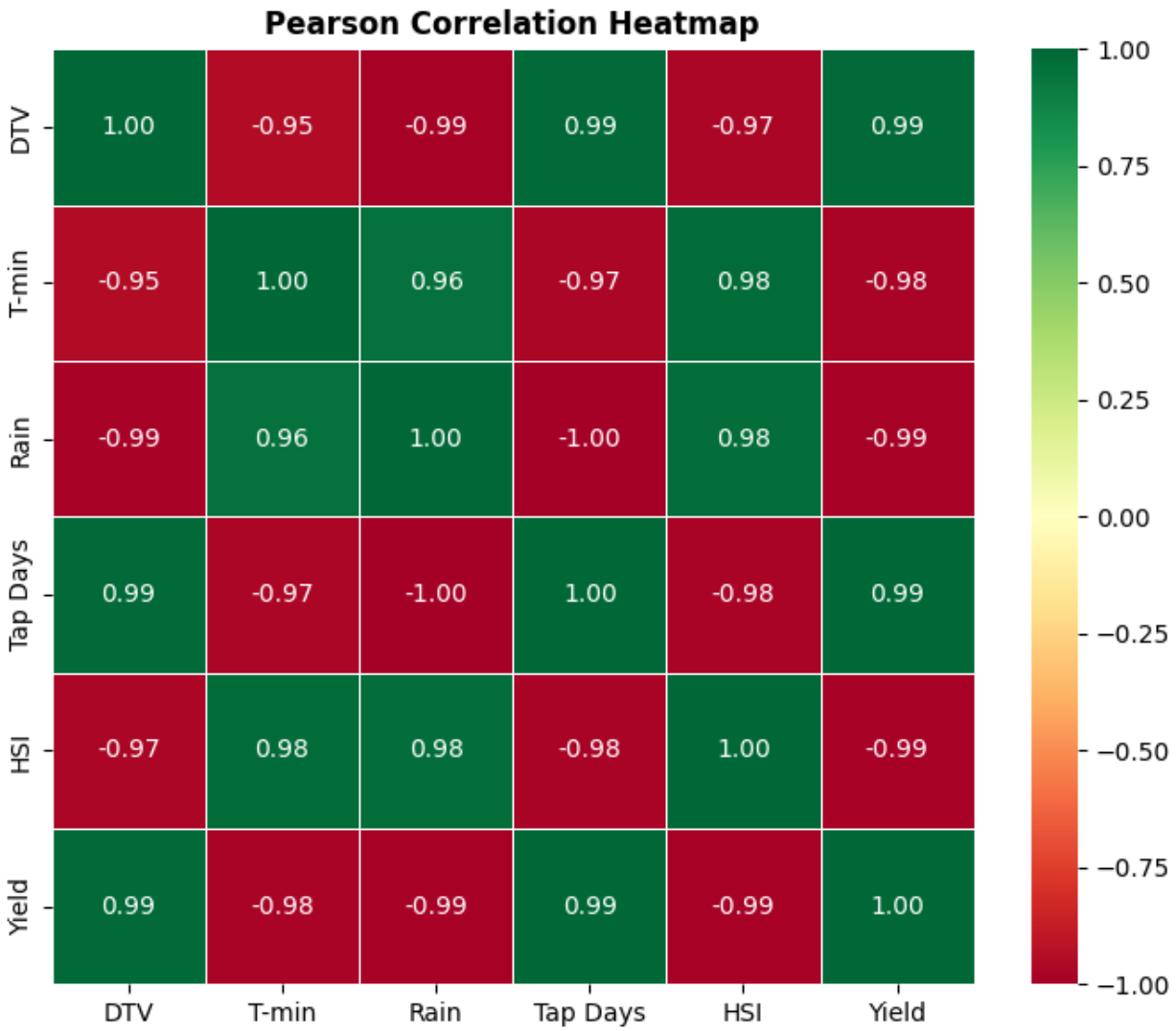
```
2021 58.9 55.0 3.9
2022 43.4 47.1 3.7
2023 46.9 46.2 0.7
Feature Importance Scores:
 HSI: 0.2349
 DTV(C): 0.2096
 Rainfall: 0.1858
 Tap Days: 0.1817
 T-min(C): 0.1745
 Year: 0.0135
```

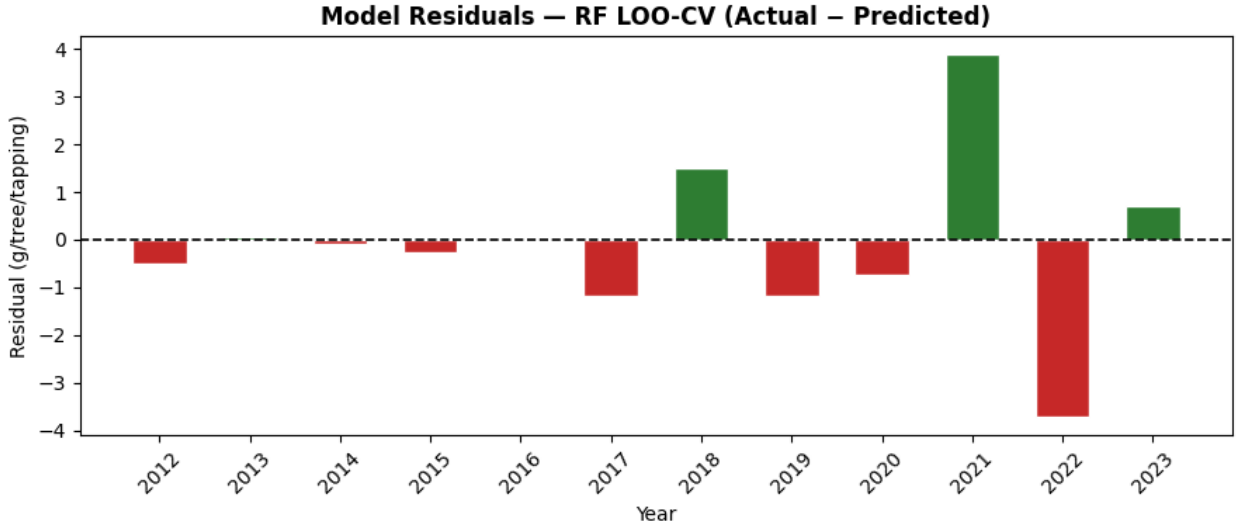





RF Feature Importance — Latex Yield Prediction
Importance Score (Gini)
0.00
0.05
0.10
0.15
0.20
0.235
0.210
0.186
0.182
0.175
0.014
HSI
DTV(C)
Rainfall
Tap Days
T-min(C)
Year


RF: Actual vs. Predicted Latex Yield (LOO-CV)
Latex Yield (g/tree/tapping)
44
46
48
50
52
54
56
58
Actual Yield
RF Predicted (LOO-CV)
Error band
R2=0.864 | MAE=1.1 | RMSE=1.7
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
Year


Pearson Correlation Heatmap
DTV
T-min
Rain
Tap Days
HSI
Yield
1.00 -0.95 -0.99 0.99 -0.97 0.99
-0.95 1.00 0.96 -0.97 0.98 -0.98
-0.99 0.96 1.00 -1.00 0.98 -0.99
0.99 -0.97 -1.00 1.00 -0.98 0.99
-0.97 0.98 0.98 -0.98 1.00 -0.99
0.99 -0.98 -0.99 0.99 -0.99 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Model Residuals — RF LOO-CV (Actual − Predicted)
Residual (g/tree/tapping)
4
3
2
1
0
−1
−2
−3
−4
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
Year